Supplementary Information for Geometric Features of Higher-Order Networks via the Spectral Triplet

Supplementary Note 1: Properties of the Laplace operator

The three operators $\mathbf{L}_\mathbf{k}^{\text{down}}$, $\mathbf{L}_\mathbf{k}^{\text{up}}$ and $\mathbf{L}_\mathbf{k}$ can be diagonalized simultaneously, and their corresponding eigenvectors obey the following relations:

$$\text{im}(\mathbf{L}_\mathbf{k}^{\text{down}}) \subseteq \ker(\mathbf{L}_\mathbf{k}^{\text{up}}),$$
$$\text{im}(\mathbf{L}_\mathbf{k}^{\text{up}}) \subseteq \ker(\mathbf{L}_\mathbf{k}^{\text{down}}).$$
$$\ker(\mathbf{L}_\mathbf{k}) = \ker(\mathbf{L}_\mathbf{k}^{\text{up}}) \cap \ker(\mathbf{L}_\mathbf{k}^{\text{down}}).$$

The eigenvectors of $\mathbf{L}^k$ are classified into the following categories:

- An eigenvector is a simultaneous eigenvector of $\mathbf{L}_k^{\text{down}}$ and $\mathbf{L}_k^{\text{up}}$ corresponding to a nonzero eigenvalue λ^k of $\mathbf{L}_k^{\text{down}}$, and a zero eigenvalue of $\mathbf{L}_k^{\text{up}}$.
- An eigenvector is a simultaneous eigenvector of $\mathbf{L}_k^{\text{down}}$ and $\mathbf{L}_k^{\text{up}}$ corresponding to a nonzero eigenvalue λ^k of $\mathbf{L}_k^{\text{up}}$, and a zero eigenvalue of $\mathbf{L}_k^{\text{down}}$.
- An eigenvector is an eigenvector common to $\mathbf{L}_k$, $\mathbf{L}_k^{\text{down}}$, and $\mathbf{L}_k^{\text{up}}$, corresponding to the eigenvalue $\lambda^k = 0$.

We provide a proof for the above in what follows. Consider an eigenvector $\boldsymbol{\xi}$ of $\mathbf{L}_k^{\text{up}}$ corresponding to $\lambda \neq 0$. This is translated as:

$$\mathbf{L}_k^{\text{up}}\boldsymbol{\xi} = \mathbf{B}_{k+1}\mathbf{B}_{k+1}^T\boldsymbol{\xi} = \lambda\boldsymbol{\xi} \quad \Rightarrow \quad \boldsymbol{\xi} = \frac{1}{\lambda}\mathbf{B}_{k+1}\mathbf{B}_{k+1}^T\boldsymbol{\xi}$$

Next, we apply $\mathbf{L}_k^{\text{down}}$, and obtain:

$$\mathbf{L}_k^{\text{down}}\boldsymbol{\xi} = \mathbf{B}_k^T\mathbf{B}_k\boldsymbol{\xi} = \frac{1}{\lambda}\mathbf{B}_k^T\mathbf{B}_k\mathbf{B}_{k+1}\mathbf{B}_{k+1}^T\boldsymbol{\xi} = \mathbf{0}$$

Supplementary Note 2: Distance: Constraints' Computations

In [**?**], the following simple equality for operators on Hilbert spaces is invoked:

$$\|T\| = \|T^*\|$$

in order to compute $|[D, f]\|$. We thus use:

$$\|[B, f]\| = \|[B, \bar{f}]\| = \|[B^T, f]\|,$$

leading to a definition of our constraint in terms of the incidence matrix (or exterior derivative in the language of discrete differential geometry):

$$\|[D, f]\| = \|[B, f]\|$$

The first constraint given by the above equation is over oriented edges $[i, j]$, which is the same as that of the graph, up to a factor of a half, whose commutator is given by:

$$[D, f]\alpha(i, j) = D(f\alpha(i, j)) - f(D\alpha(i, j)),$$

$f \in \Omega^0$ is 0-form (scalar) and $\alpha \in \Omega^1$ is 1-form (co-vector or in simple terms a row vector as opposed to a column vector). Their multiplication should give $\Omega^1 \times \Omega_0 \to \Omega^1$, which means that f has to be pulled up to the dimension of α through an averaging given by:

$$(f\alpha)([i, j]) = (F_0\alpha)([i, j]) = \frac{f_i + f_j}{2}\alpha([i, j]),$$

which is used to find the commutator. This gives:

$$B_1(F_1\alpha(i, j)) = \frac{(f_i + f_j)}{2}(\alpha(j) - \alpha(i)),$$

and

$$F_0(B_1\alpha(i, j)) = F_0(\alpha(j) - \alpha(i)) = f_j\alpha(j) - f_i\alpha(i).$$

Then the commutator is:

$$[D, f]\alpha(i, j) = \frac{(f_i + f_j)}{2}(\alpha(j) - \alpha(i)) - (f_j\alpha(j) - f_i\alpha(i))$$

$$= \frac{f_i}{2}\alpha(j) - \frac{f_i}{2}\alpha(i) + \frac{f_j}{2}\alpha(j) - \frac{f_j}{2}\alpha(i) - f_j\alpha(j) + f_i\alpha(i)$$

$$[D, f]\alpha(i, j) = \frac{(f_i - f_j)}{2}(\alpha(i) + \alpha(j))$$

Similarly to what we did for an edge, for an oriented triangle, $[i, j, k]$, f has to be pulled up to the dimension of $\omega \in \Omega^2$, the 2-form on the triangle, through an averaging given by:

$$(f\omega)([i, j, k]) = \frac{f_i + f_j + f_k}{3}\omega([i, j, k]),$$

The commutator is then given by:

$$[D, f] = \begin{pmatrix} 0 & B_1F_1 - F_0B_1 & 0 & 0 \\ B_1^\top F_0 - F_1B_1^\top & 0 & B_2F_2 - F_1B_2 & 0 \\ 0 & B_2^\top F_1 - F_2B_2^\top & 0 & B_3F_3 - F_2B_3 \\ 0 & 0 & B_3^\top F_2 - F_3B_3^\top & 0 \end{pmatrix}$$

In this case, the boundary operator $B_2 : \Omega^2 \to \Omega^1$ acts on the triangle:

$$B_2(\omega)([i, j, k]) = \omega([j, k]) - \omega([i, k]) + \omega([i, j]).$$

And,

$$\begin{aligned} f(B_2\omega)([i, j, k]) &= f(\omega([j, k]) - \omega([i, k]) + \omega([i, j])) = F_1(\omega([j, k]) - \omega([i, k]) + \omega([i, j])) \\ &= \frac{f_j + f_k}{2}\omega([j, k]) - \frac{f_i + f_k}{2}\omega([i, k]) + \frac{f_i + f_j}{2}\omega([i, j]). \end{aligned}$$

$$B_2(f\omega)([i, j, k]) = B_2(F_2\omega)([i, j, k]) = \frac{f_i + f_j + f_k}{3}B_2\omega([i, j, k]) = \frac{f_i + f_j + f_k}{3}[\omega([j, k]) - \omega([i, k]) + \omega([i, j])].$$

Then the commutator is given by:

$$[B_2, f]\omega([i,j,k]) = \frac{f_j + f_k - 2f_i}{6}\omega([j,k]) + \frac{-f_i + 2f_j - f_k}{6}\omega([i,k]) + \frac{f_i + f_j - 2f_k}{6}\omega([i,j]). \tag{1}$$

The commutator $[D, f]$ contains the block $fB_2 - B_2 f = [B_2, f]$ and $fB_3 - B_3 f = [B_2, f]$. The calculation above shows that they are both $\neq 0$ only when we extend f to higher-order simplices; otherwise their to the norm $\|[D, F]\|$ vanish.

Finally, for an oriented tetrahedron $[i,j,k,l]$, f has to be pulled to the dimension of $\beta \in \Omega^3$ through an averaging given by:

$$(f\beta)([i,j,k,l]) = (F_3\beta)([i,j,k,l]) = \frac{f_i + f_j + f_k + f_l}{4}\beta([i,j,k,l]),$$

The corresponding entry in the commutator is:

$$\begin{aligned}(B_3F_3 - F_2B_3)\beta([i,j,k,l]) = \frac{1}{12}\Big[&(3f_l - f_i - f_j - f_k)\,[i,j,k] \\ &-(3f_k - f_i - f_j - f_l)\,[i,j,l] \\ &+(3f_j - f_i - f_k - f_l)\,[i,k,l] \\ &-(3f_i - f_j - f_k - f_l)\,[j,k,l]\Big]\,\beta([i,j,k,l]).\end{aligned} \tag{2}$$

Supplementary Note 3: Heat Kernel Expansion

In order to determine $m_i^{[n]}$ we varied the number of terms in the expansion and computed the quality of the fit using the Adjusted-R^2. We empirically observed that the best fit corresponded to the one where $m_i^{[n]}$ matched the maximum degree. More precisely, when we were fitting for node curvature the upper cut-off turned out to be the maximum node degree.

Supplementary Note 4: Heat Kernel Rescaling

If $L'_s = L_s/\Lambda^2$, then the transformation $L_s \to L_s/\Lambda^2$ is physically equivalent to the transformation $t \to t\Lambda^2$ leading to:

$$K(i,i,t) \sim \frac{1}{(4\pi t\Lambda^2)^{d_s/2}} \sum_{m=0}^{\infty} u_m(x,y)\Lambda^{2m} t^m,$$

$$K(i,i,t) \sim \frac{1}{(4\pi t)^{d_s/2}} \sum_{k=0}^{\infty} u_m(x,y)\Lambda^{2m-d_s} t^m,$$

For this to look like a standard heat kernel expansion for L' with coefficients u'_m, we must define:

$$u'_m(x) \equiv u_m(x) \cdot \Lambda^{2m-d_s}$$

Supplementary Note 5: Λ computation

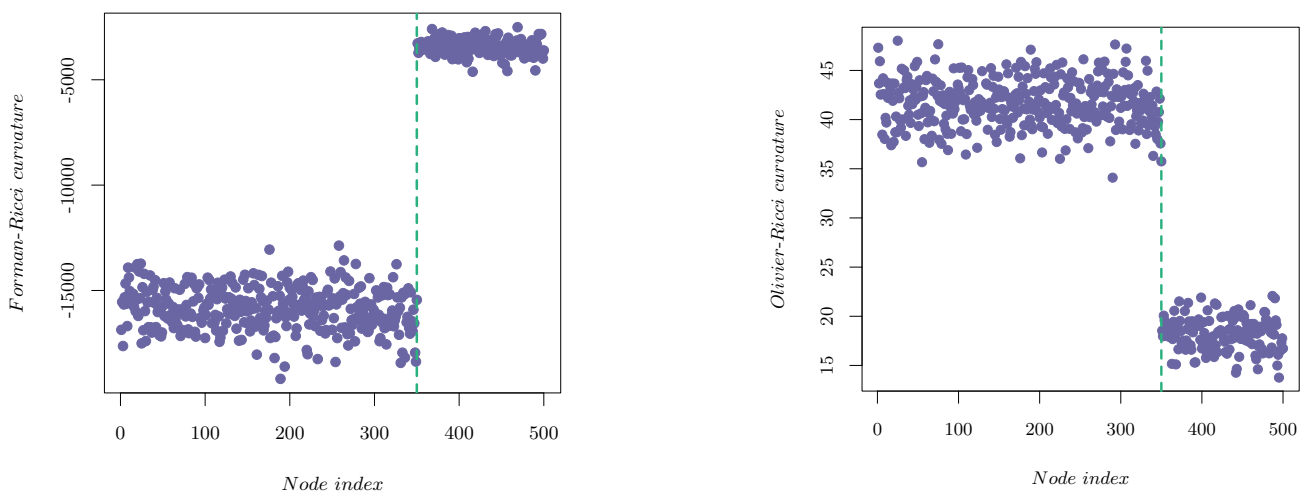
The values Λ, the smallest non-zero eigenvalue of the movement's super Lapalcian, evaluated at the end of the piece, are tabulated below.

Table S1: Lambda (Λ) values for the movements.

Movement	$\mathbf{\Lambda}$
Partita 1 Allemanda	0.697
Partita 1 Bourrée	0.089
Partita 1 Sarabande	0.253
Partita 2 Sarabande	0.246
Partita 3 Loure	0.312
Partita 3 Minuet I, II	0.950
Sonata 1 Adagio	0.327
Sonata 1 Fugue	0.045
Sonata 1 Siciliano	0.285
Sonata 2 Grave	0.272
Sonata 2 Fugue	0.050
Sonata 2 Andante	0.226
Sonata 3 Adagio	0.114
Sonata 3 Fugue	0.167
Sonata 3 Largo	0.765

Supplementary Note 6: Forman and Oliver node curvature of the SBM.

Here we show the Forman and Oliver node curvatures for the SBM graph.

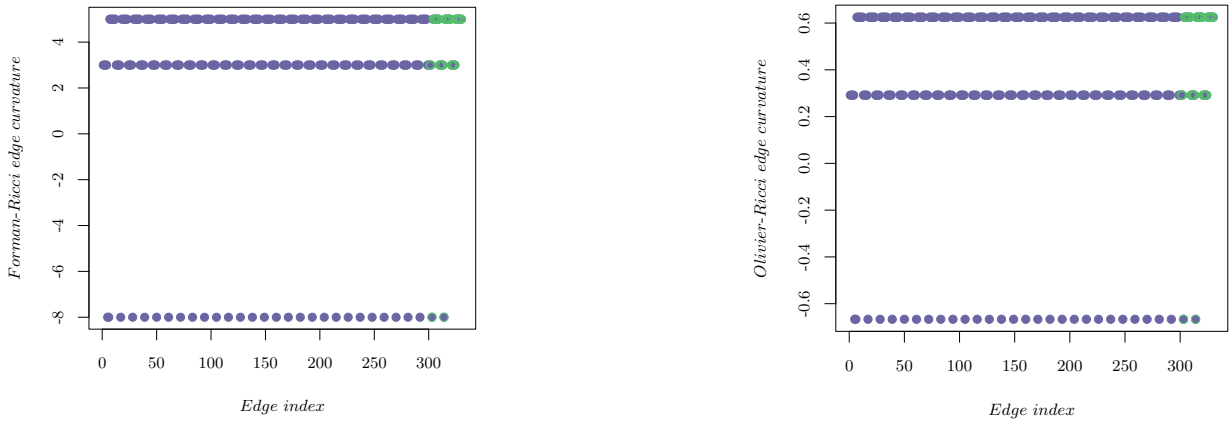


(a) (b)

Figure S1: The Forman and Olivier-Ricci curvature of the graph in Figure 4.

Supplementary Note 7: Forman and Oliver node curvature of the ring of cliques.

Here we show the Forman and Oliver node curvatures for the ring of cliques graph.

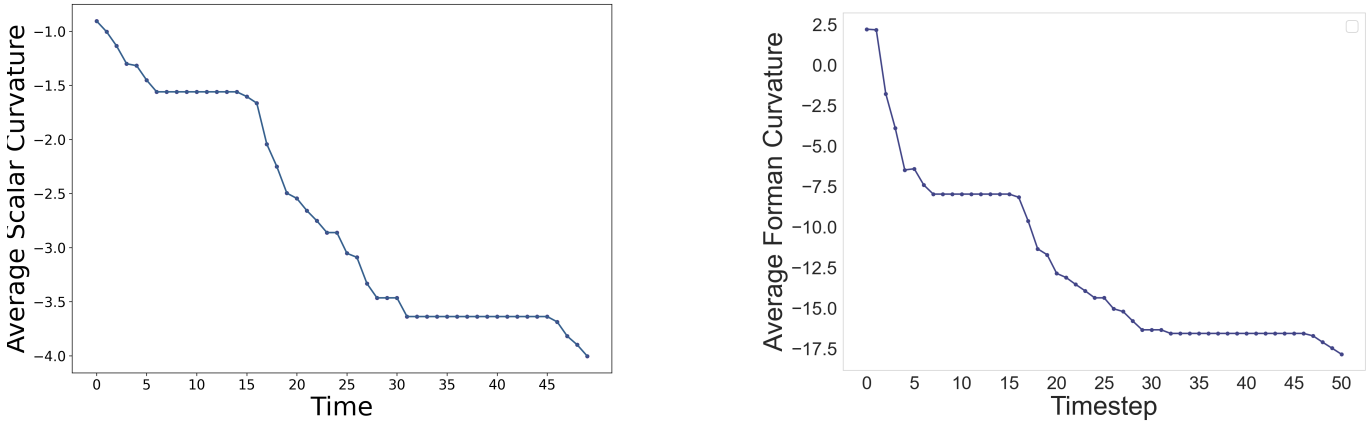


(a) (b)

Figure S2: Both Forman and Olivier-Ricci edge curvatures are shown in S2a and S2b respectively, which clearly fail to distinguish the orange edges of the graph, whose curvature values are highlighted in green.

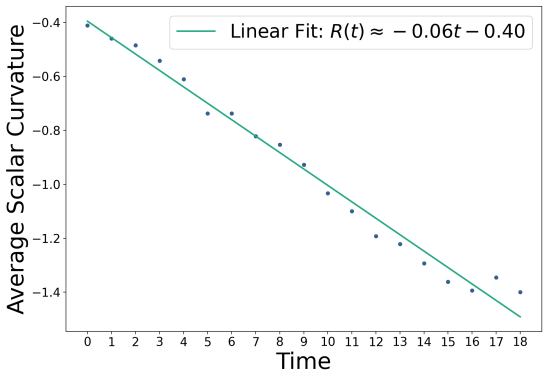
Supplementary Note 8: Temporal evolution of average scalar curvature

Here we show the evolution of a sample of average scalar curvatures compared with the Forman counterpart.

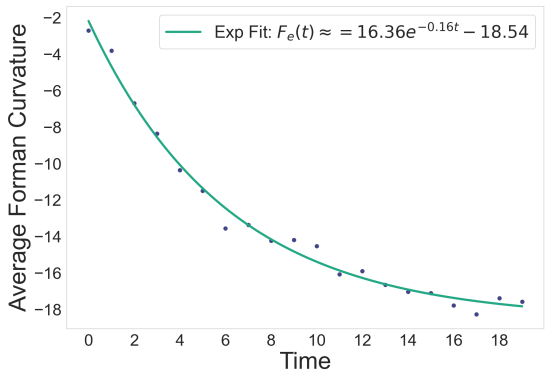


(a) (b)

Figure S3: The evolution of the average scalar curvature using our proposed method is shown in the figure on the left, while the evolution of the average Forman-Ricci node curvature is shown in the figure on the right for the dance movement Sarabande from Partita 2.



(a)



(b)

Figure S4: The evolution of the average scalar curvature using our proposed method is shown in the figure on the left, while the evolution of the average Forman-Ricci node curvature is shown in the figure on the right for the slow movement Largo from Sonata 3.

Supplementary Note 9: Connes distance in musical spaces

We show below the results for rest of the movements for the recovered $d_{\mathcal{N},C}$.

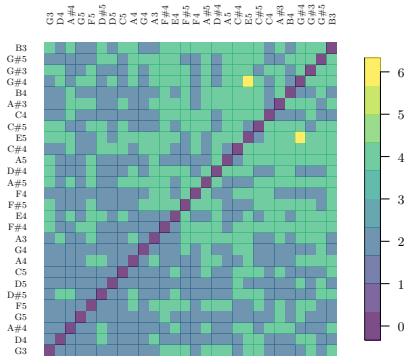


Figure S5: Sonata 1 Adagio

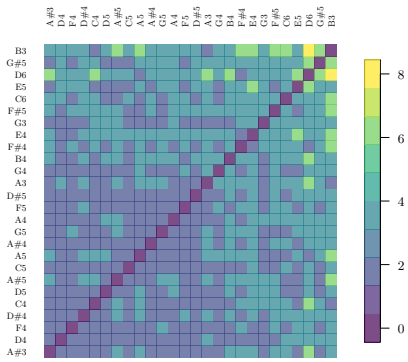


Figure S6: Sonata 1 Siciliano

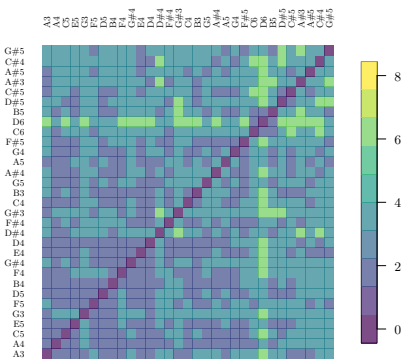


Figure S7: Sonata 2 Grave

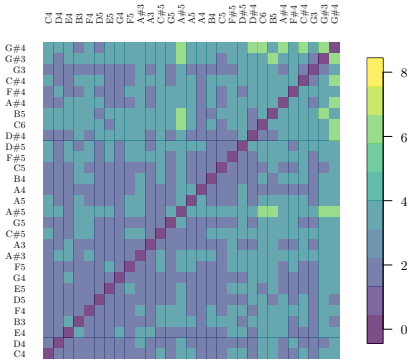


Figure S8: Sonata 3 Adagio

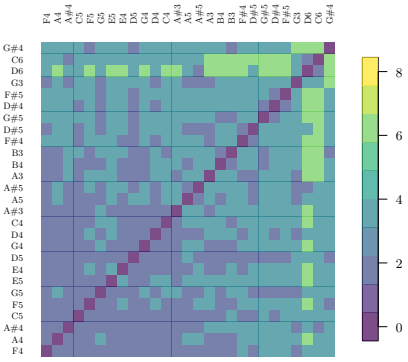


Figure S9: Sonata 3 Largo

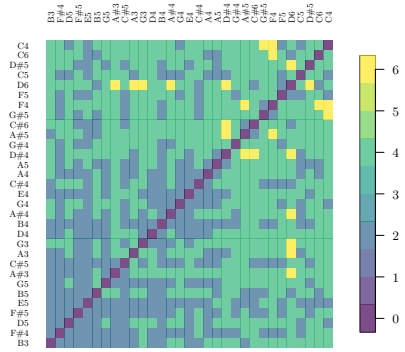


Figure S10: Partita 1 Allemanda

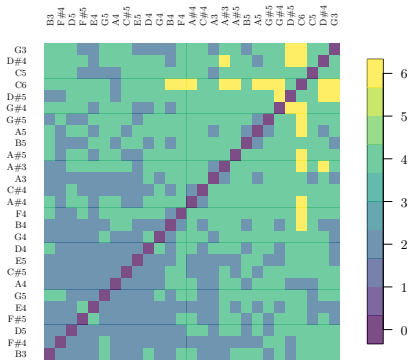


Figure S11: Partita 1 Sarabande

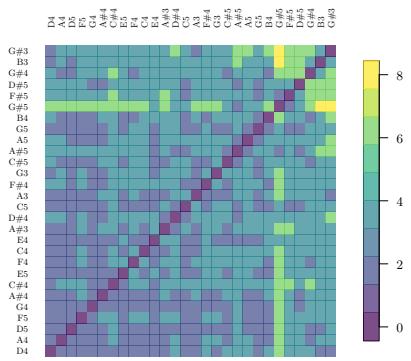


Figure S12: Partita 2 Sarabande

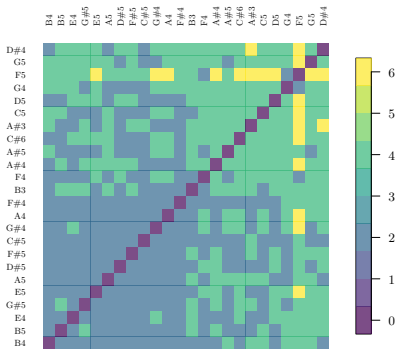


Figure S13: Partita 3 Loure

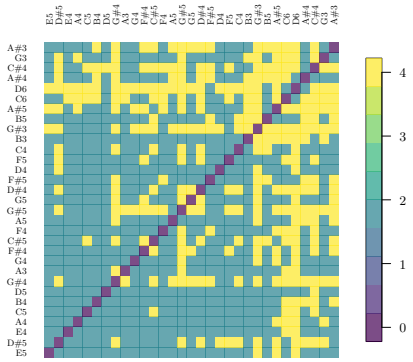


Figure S14: Sonata 2 Fugue

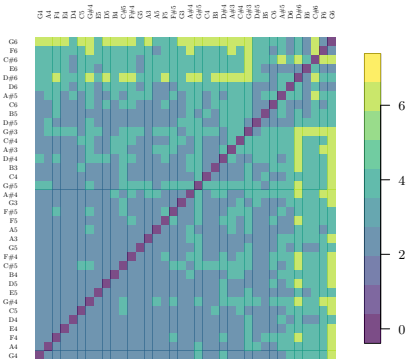


Figure S15: Sonata 3 Fugue

Supplementary Note 10: Pairwise distance

We show the pairwise distances between notes defined as the geodesic over the simple graph using `distances` of `igraph`, which does not use the Connes distance. It's a Dijkstra algorithm over the graph.

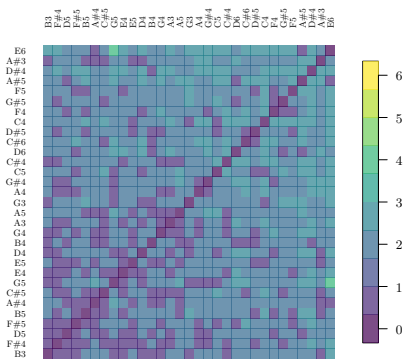


Figure S16: Partita 1 Bourrée

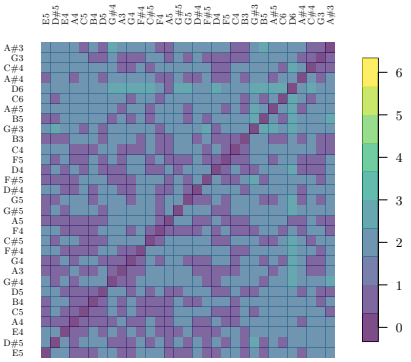


Figure S17: Sonata 2 Fugue

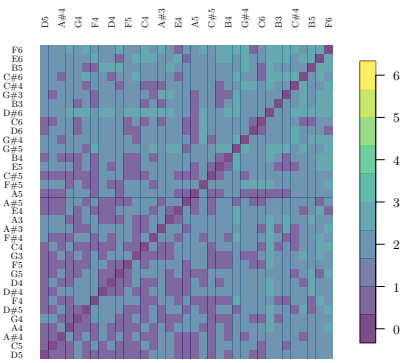


Figure S18: Sonata 1 Fugue